EXERCISE 02 - SQL FOUNDATIONS
SQL AGGREGATE FUNCTIONS & OPERATORS

Question 1

```
SELECT DISTINCT department
FROM students;
```

Department
IT
HR
FINANCE

Question 2

```
SELECT department,
        AVG(age) AS avg_age
FROM students
GROUP BY department;
```

department	avg, age
IT	20,5
HR	22
Finance	23

Question 3

```
SELECT department,
        COUNT(*) AS student_count
FROM students
GROUP BY department
HAVING COUNT(*) > 1;
```

Department	student_count
IT	2
HR	2

Question 4

```
SELECT student_id,
       name,
       age,
       department
FROM students
WHERE age
BETWEEN 21 AND 23;
```

Student_id	Name	Age	department
3	Charlie	21	IT
2	Bob	22	HR
5	Eve	22	HR
4	Diana	23	Finance

Question 5

```
SELECT student_id,
       name,
       age,
       department
FROM students
WHERE department IN('IT', 'HR')
 AND age > 21;
```

Student_id	Name	Age	Department
2	Bob	22	HR
5	Eve	22	HR

02 The Courses Table

QUESTION 6

```
SELECT department, SUM(credits) AS total_credits
FROM courses
GROUP BY department
HAVING SUM(credits) > 5;
```

department	total_credits
IT	11

Question 7

```
SELECT course_id,
       course_name,
       department,
       credits
FROM courses
WHERE credits NOT 4;
```

course_id	course_name	department	Credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

Question 8

```
SELECT course_id,
       course_name,
       credits
FROM courses
ORDER BY credits DESC
LIMIT 3;
```

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

~~Question 9~~

03 ENROLLMENTS TABLE

Question 9

```
SELECT MAX(grade) AS max_grade,
       Min(grade) AS min_grade,
       AVG(grade) AS avg_grade
From enrollments;
```

max_grade	min_grade	avg_grade
90	78	84,6

Question 10

```
SELECT course_id,
       COUNT(*) AS enrollment_count
FROM enrollments
GROUP BY course_id;
```

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

04 SALARIES TABLE

Question 11

```
SELECT department,
       sum(salary) AS total_salary,
       SUM(bonus) AS total_bonus
FROM salaries
GROUP BY department;
```

department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

Question 12

```
SELECT department,
       AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG(salary) > 55000;
```

department	avg_salary
IT	61000
Finance	70000

Question 13

```
SELECT employee_id,
       name,
       salary,
       bonus,
       (salary + bonus) AS total_compensation
FROM salaries
WHERE (salary + bonus) > 60000;
```

employee_id	name	Salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

05 PROJECTS TABLE

Question 14

```
SELECT department,
       SUM(budget) AS total_budget,
       AVG(budget) AS avg_budget
FROM projects
GROUP BY department
HAVING AVG(budget) > 70000;
```

department	total_budget	avg_budget
IT	270 000	135 000
Finance	80000	80 000

Question 15

```
SELECT project_id,
       project_name,
       department,
       budget
FROM projects
WHERE budget BETWEEN 50000 AND 120000
AND department NOT 'Marketing';
```

Project_id	Project_name	department	budget
1	AI App	IT	120000
2	PayrollSystem	Finance	80000
5	HR Portal	HR	50000